

MAJOR PROJECT REGISTRATION FORM

Date: 31-07-2026

Project Category:	☒ Research oriented	☐ Application oriented	Batch ID:	C7
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: **Research Paper Summarization and Gap Identification Using Transformers**

Abstract

Research papers are a primary source of knowledge and innovation in academia and industry. However, the rapid increase in research publications makes it difficult for researchers to efficiently review large volumes of literature. Researchers often spend significant time reading multiple papers to understand existing solutions, identify limitations, discover research gaps, and explore future research directions. This challenge particularly affects research scholars, postgraduate students, final-year project students, academicians, and industry researchers who regularly conduct literature reviews. Currently, researchers rely on manual analysis and AI tools such as ChatGPT and Gemini for research paper summarization. While these tools generate concise summaries, they mainly focus on individual papers and provide limited support for research gap identification, limitation extraction, future work analysis, and comparison of multiple papers. As a result, researchers still need to invest considerable effort in identifying unexplored research opportunities.

To overcome these challenges, recent advancements in **Natural Language Processing (NLP) and Deep Learning** have provided new possibilities for automated research paper analysis. Transformer-based models such as T5, BART, and PEGASUS have demonstrated remarkable performance in abstractive text summarization by generating coherent and context-aware summaries. These models have been successfully applied in various domains for document understanding and information extraction.

However, existing systems primarily focus on summary generation and lack comprehensive research analysis capabilities such as research gap identification, limitation extraction, future work analysis, and comparative study of multiple research papers. To address this limitation, the proposed project, **Research Paper Summarization and Gap Identification Using Transformers**, generates structured summaries, automatically identifies research gaps, extracts limitations, suggests future research directions, and performs comparative analysis of multiple papers on the same topic.

The system also evaluates the performance of T5, BART, and PEGASUS models to determine the most effective approach for research paper analysis. The project focuses on English-language research papers and aims to assist researchers in conducting efficient literature reviews and identifying potential research opportunities. The expected outcome is an intelligent research assistance system that reduces literature review effort, improves research understanding, and provides meaningful insights for future research.

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References (Category specific)

1. A Comparative Study of PEGASUS, BART, and T5 for Text Summarization Across Diverse Datasets
Link: <https://www.mdpi.com/1999-5903/17/9/389>
2. Multi-Document Text Summarization Using Ensemble of Attention-Based Transformer Models
Link: <https://doi.org/10.5220/0014207300004932>
3. Evaluating LLMs and Pre-trained Models for Text Summarization Across Diverse Datasets
Link: <https://arxiv.org/abs/2502.19339>
4. Summarization from Leaderboards to Practice: Choosing a Representation Backbone and Ensuring Robustness
Link: <https://arxiv.org/abs/2306.10555>
5. An Analysis of Abstractive Text Summarization Using Pre-trained Models
Link: <https://arxiv.org/abs/2303.12796>

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